Binary Phase Shift Keying

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1 introduction

Binary phase shift keying (BPSK) is using a binary input to change the phase of a signal for modulation purposes.

2 Data and analysis

Figure 1 shows the original signal and figure 2 shows the BPSK signal.

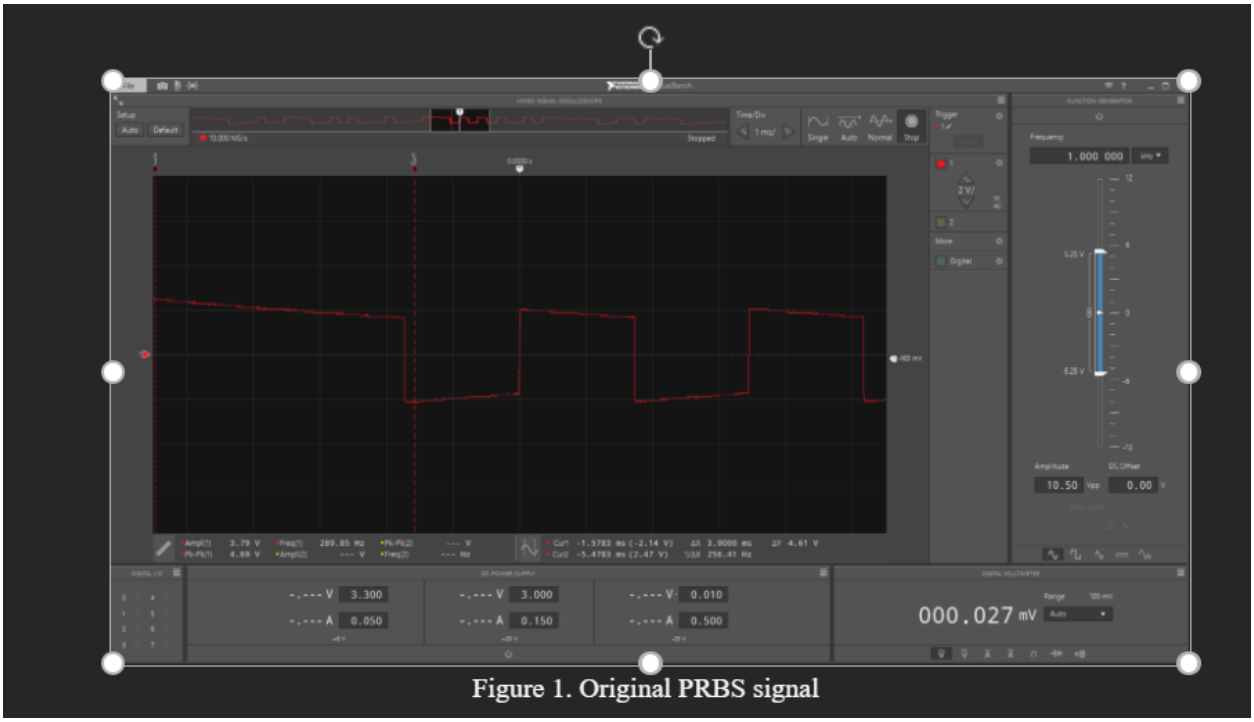


Figure 1: Original signal

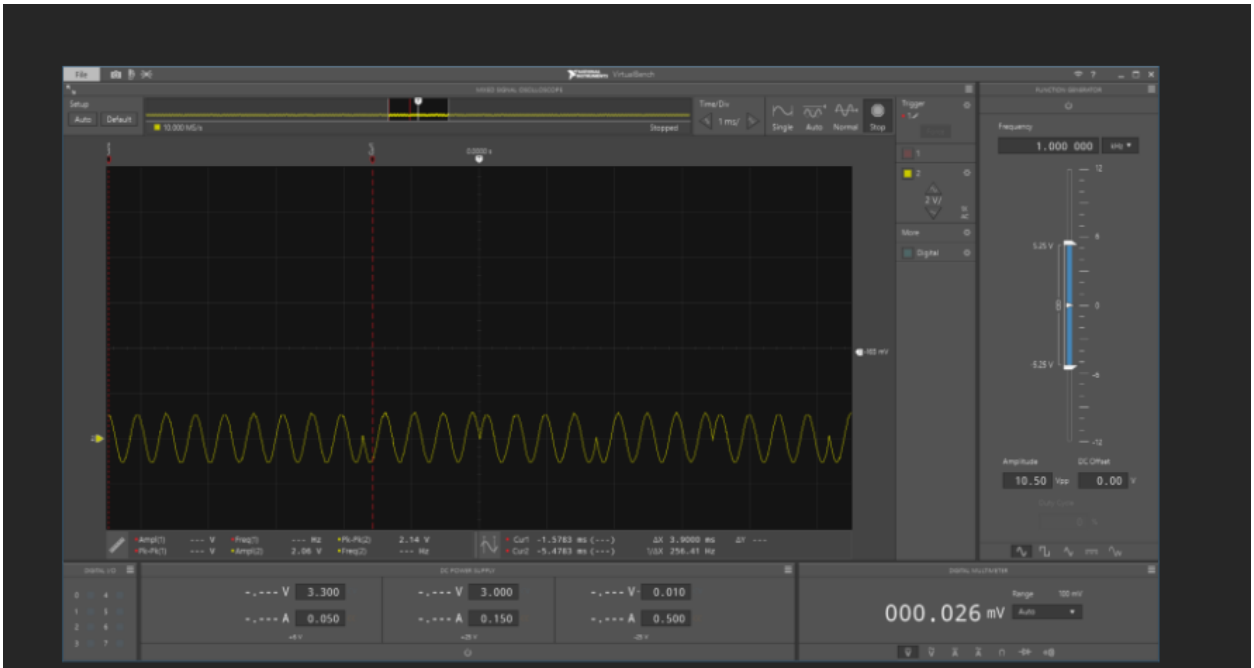


Figure 2: modulated signal

As seen in figure 2, the phase of the signal changes at the rising and falling edges of the binary sequence. Figure 3 below shows the results of the decision maker circuit compared to the original PRBS and figure 4 is the demodulated signal compared to the original.

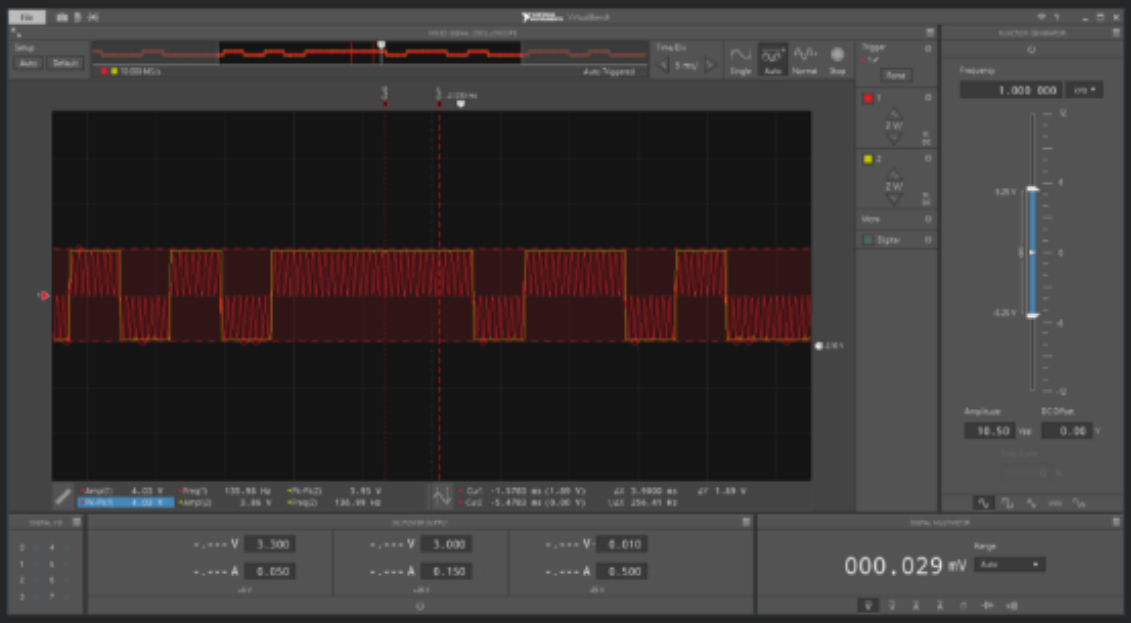


Figure 3. Decision Maker Output Compared to Original PRBS



Figure 4. Line-Code Decoder Compared to Original PRBS

Certification: **I hereby certify that performance of the experiment above and the writing of this report was entirely by me (or by me and my lab partner). I understand that if this certi fication is false, I am in violation of Academic Honesty rules and may be subject to serious penalty in accordance with those rules.**

Signature Josiah Germain **Date** 11/20/25